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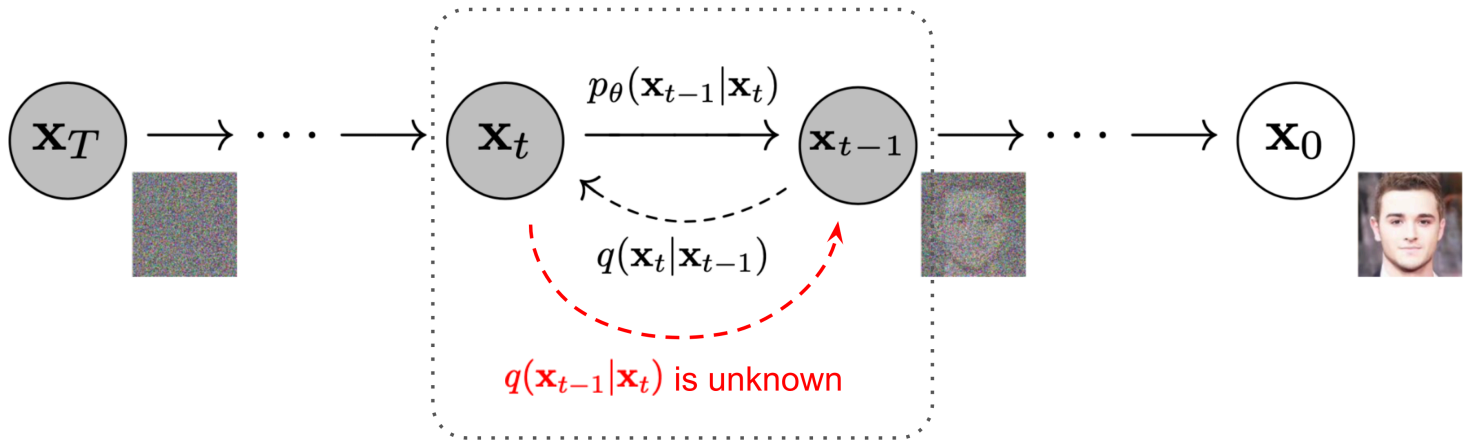
Deep Generative Models

Lecture 8

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Recap of Previous Lecture



Reverse Process (Ancestral Sampling)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)}$$
$$\approx p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

If β_t is sufficiently small, $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is approximately Gaussian.

Recap of Previous Lecture

Forward Process

1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. $\mathbf{x}_t = \sqrt{1 - \beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t$
3. $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$

Reverse Process

1. $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$
2. $\mathbf{x}_{t-1} = \boldsymbol{\sigma}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\boldsymbol{\epsilon} + \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t)$
3. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$

SNR and Log-SNR

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

$$\text{SNR}(t) = \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t}.$$

$$\lambda_t = \log \text{SNR}(t), \qquad \bar{\alpha}_t = \frac{1}{1 + e^{-\lambda_t}}.$$

$\lambda_t > 0$: signal dominates; $\lambda_t = 0$: equal scales; $\lambda_t < 0$: noise dominates.

Weng L. What are Diffusion Models?, blog post, 2021
Kingma D. et al. Variational Diffusion Models, 2021

Recap of Previous Lecture

Forward process maps any distribution $p_{\text{data}}(\mathbf{x})$ to $\mathcal{N}(0, \mathbf{I})$ by injection of noise:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\,\mathbf{x}_{t-1}, \beta_t\,\mathbf{I});$$
$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\,\mathbf{I}).$$

Reverse process refers to an intractable distribution that can be approximated by a normal distribution (with unknown parameters) for small β_t:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx \mathcal{N}\left(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right)$$

Conditioned reverse process is a normal distribution with known parameters, describing how to denoise a noisy image $\mathbf{x}_t$ when we know the clean image $\mathbf{x}_0$.

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t\,\mathbf{I})$$

Recap of Previous Lecture

- $\mathbf{z} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$ represents the latent variables.
- Variational posterior distribution:

$$q(\mathbf{z}|\mathbf{x}) = q(\mathbf{x}_1, \ldots, \mathbf{x}_T|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}).$$

- Generative model and prior:

$$p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1); \quad p_{\boldsymbol{\theta}}(\mathbf{z}) = \prod_{t=2}^{T} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) \cdot p(\mathbf{x}_T)$$

Das A. An Introduction to Diffusion Probabilistic Models, blog post, 2021
Ho J. Denoising Diffusion Probabilistic Models, 2020

Recap of Previous Lecture

Standard ELBO

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) \geq \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z}|\mathbf{x})} = \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \to \max_{\boldsymbol{\phi},\boldsymbol{\theta}}$$

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)\|p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$

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Recap of Previous Lecture

Diffusion Parameterizations

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}.$$

Network output at $\mathbf{x}_t$	Regression target	Clean-data estimate
Noise $\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}$	$\boldsymbol{\epsilon}$	$\dfrac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}}{\sqrt{\bar{\alpha}_t}}$
Clean data $\hat{\mathbf{x}}_{\boldsymbol{\theta},t}$	$\mathbf{x}_0$	$\hat{\mathbf{x}}_{\boldsymbol{\theta},t}$
Score $\mathbf{s}_{\boldsymbol{\theta},t}$	$-\dfrac{\boldsymbol{\epsilon}}{\sqrt{1-\bar{\alpha}_t}}$	$\dfrac{\mathbf{x}_t + (1-\bar{\alpha}_t)\,\mathbf{s}_{\boldsymbol{\theta},t}}{\sqrt{\bar{\alpha}_t}}$
v-prediction $\mathbf{v}_{\boldsymbol{\theta},t}$	$\sqrt{\bar{\alpha}_t}\,\boldsymbol{\epsilon} - \sqrt{1-\bar{\alpha}_t}\,\mathbf{x}_0$	$\sqrt{\bar{\alpha}_t}\,\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\mathbf{v}_{\boldsymbol{\theta},t}$

Convertible predictions; different losses. Plain MSE assigns different weights to different noise levels.

Salimans T., Ho J. Progressive Distillation for Fast Sampling of Diffusion Models, 2022

Recap of Previous Lecture

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L}_{\text{simple}} = \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}, t}(\mathbf{x}_t)\|^2$.

Sampling (Ancestral)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute the mean of $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t), \tilde{\beta}_t \cdot \mathbf{I})$:

$$\boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t (1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta}, t}(\mathbf{x}_t).$$

3. Denoise $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.

Outline

01 DDPM as a Score-Based Generative Model

02 Model Guidance

Classifier Guidance
Classifier-Free Guidance

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01 **DDPM as a Score-Based Generative Model**

02 Model Guidance

Classifier Guidance
Classifier-Free Guidance

Denoising Diffusion as a Score-Based Generative Model

DDPM Objective

$$\mathcal{L}_t = \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})} \left[C_{1,t} \cdot \left\| \boldsymbol{\epsilon}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) - \boldsymbol{\epsilon} \right\|_2^2 \right]$$

Denoising Diffusion as a Score-Based Generative Model

DDPM Objective

$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[C_{1,t}\cdot\left\|\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)-\boldsymbol{\epsilon}\right\|_2^2\right] \\ &= \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[C_{2,t}\cdot\left\|\frac{\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)}{\sqrt{1-\bar{\alpha}_t}}-\frac{\boldsymbol{\epsilon}}{\sqrt{1-\bar{\alpha}_t}}\right\|_2^2\right]\end{aligned}$$

Denoising Diffusion as a Score-Based Generative Model

DDPM Objective

$$\mathcal{L}_t = \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[C_{1,t}\cdot\left\|\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)-\boldsymbol{\epsilon}\right\|_2^2\right]$$

$$= \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[C_{2,t}\cdot\left\|\frac{\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)}{\sqrt{1-\bar{\alpha}_t}}-\frac{\boldsymbol{\epsilon}}{\sqrt{1-\bar{\alpha}_t}}\right\|_2^2\right]$$

$$q(\mathbf{x}_t|\mathbf{x}_0)=\mathcal{N}(\sqrt{\bar{\alpha}_t}\cdot\mathbf{x}_0,(1-\bar{\alpha}_t)\cdot\mathbf{I})$$

Denoising Diffusion as a Score-Based Generative Model

DDPM Objective

$$\mathcal{L}_t = \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[C_{1,t}\cdot\left\|\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)-\boldsymbol{\epsilon}\right\|_2^2\right]$$

$$= \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[C_{2,t}\cdot\left\|\frac{\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)}{\sqrt{1-\bar{\alpha}_t}}-\frac{\boldsymbol{\epsilon}}{\sqrt{1-\bar{\alpha}_t}}\right\|_2^2\right]$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\cdot\mathbf{x}_0, (1-\bar{\alpha}_t)\cdot\mathbf{I})$$

$$\nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t-\sqrt{\bar{\alpha}_t}\cdot\mathbf{x}_0}{1-\bar{\alpha}_t} = -\frac{\boldsymbol{\epsilon}}{\sqrt{1-\bar{\alpha}_t}}.$$

Denoising Diffusion as a Score-Based Generative Model

We can reparameterize the model as:

$$\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) = -\frac{\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)}{\sqrt{1-\bar{\alpha}_t}} = \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t).$$

This is the noise-to-score identity from **Tweedie's formula (Lecture 6)** at the MSE optimum.

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$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left[C_{2,t} \cdot \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) \right\|_2^2\right]$$

DDPM vs NCSN: Objectives

DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left[C_{2,t}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2\right]$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\cdot\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\cdot\boldsymbol{\epsilon}$$

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In practice, this coefficient is often omitted.

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NCSN Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2$$

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t\cdot\boldsymbol{\epsilon}$$

DDPM vs NCSN: Objectives

DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left[C_{2,t}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2\right]$$

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$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t\cdot\boldsymbol{\epsilon}$$

Maximizing the ELBO leads to the same objective as denoising score matching!

DDPM and NCSN: Connecting Noise Coordinates

DDPM scales the clean signal as it adds noise:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

Song J., Meng C., Ermon S. Denoising Diffusion Implicit Models, 2021
Song Y. et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

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Divide by the signal coefficient

$$\underbrace{\frac{\mathbf{x}_t}{\sqrt{\bar{\alpha}_t}}}_{\tilde{\mathbf{x}}_t} = \mathbf{x}_0 + \underbrace{\sqrt{\frac{1-\bar{\alpha}_t}{\bar{\alpha}_t}}}_{\sigma_t}\,\boldsymbol{\epsilon}.$$

Here σ_t measures additive noise in the rescaled state.

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Here σ_t measures additive noise in the rescaled state.

Recover the NCSN noise form

$$\tilde{\mathbf{x}}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}, \qquad q(\tilde{\mathbf{x}}_t \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2 \mathbf{I}).$$

As $\bar{\alpha}_t$ decreases, the noise level σ_t increases.

The Gaussian perturbations are connected by rescaling. Next, compare their sampling rules: ancestral updates and Langevin dynamics.

Song J., Meng C., Ermon S. Denoising Diffusion Implicit Models, 2021
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DDPM vs NCSN: Sampling

Sampling (Ancestral)

$$\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$$

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}$$

DDPM vs NCSN: Sampling

Sampling (Ancestral)

$$\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$$

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}$$

$$= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}$$

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Sampling (Ancestral)

$$\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$$

$$\begin{aligned}\mathbf{x}_{t-1} &= \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon} \\ &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{\alpha_t(1-\bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon} \\ &= \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}\end{aligned}$$

DDPM vs NCSN: Sampling

Sampling (Annealed Langevin Dynamics)

1. Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
2. Update $\mathbf{x}_t^l$ via L steps of Langevin dynamics at each noise level σ_t:

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \cdot \boldsymbol{\epsilon}_t^l.$$

3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to the next noise level σ_{t-1}.

DDPM vs NCSN: Summary

Summary

- Gaussian perturbations:
  - DDPM: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$;
  - NCSN: $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \boldsymbol{\epsilon}$.
  - Rescaling the DDPM state by $1/\sqrt{\bar{\alpha}_t}$ gives the NCSN noise form.

Kingma D. et al. Variational Diffusion Models, 2021
Song Y. et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

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  - Rescaling the DDPM state by $1/\sqrt{\bar{\alpha}_t}$ gives the NCSN noise form.
- The objectives coincide: ELBO $\equiv$ score-matching.

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Summary

- Gaussian perturbations:
  - DDPM: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$;
  - NCSN: $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \boldsymbol{\epsilon}$.
  - Rescaling the DDPM state by $1/\sqrt{\bar{\alpha}_t}$ gives the NCSN noise form.
- The objectives coincide: ELBO $\equiv$ score-matching.
- The sampling procedures differ:
  - Ancestral sampling in DDPM;
  - Annealed Langevin dynamics for NCSN;
  - Hybrid approaches that combine both updates are possible.

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DDIM: From Noise to Clean Data

Can we change the sampler without retraining the DDPM network?

Keep fixed: $q(\mathbf{x}_t \mid \mathbf{x}_0)$ at every noise level — the same noise-MSE training.

Change: the joint $q(\mathbf{x}_{1:T} \mid \mathbf{x}_0)$ and its transitions, using a non-Markovian forward process.

At each noise level, the forward process gives:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}.$$

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At each noise level, the forward process gives:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}.$$

Predict the noise with $\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)$ and solve for the clean data:

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)}{\sqrt{\bar{\alpha}_t}}.$$

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Predict the noise with $\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)$ and solve for the clean data:

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)}{\sqrt{\bar{\alpha}_t}}.$$

These two estimates exactly reconstruct the current state:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t).$$

DDIM idea: keep both estimates fixed for one transition and change only the weights, from t to $s < t$.

DDIM: Same Marginals, Different Transitions

Let $X \sim \mathcal{N}(0, 4)$ and $Z \sim \mathcal{N}(0, 1)$ be independent. Target: $Y \sim \mathcal{N}(0, 1)$.

Stochastic transition

$$Y = \frac{X}{4} + \frac{\sqrt{3}}{2} Z.$$

Variance: $\frac{4}{16} + \frac{3}{4} = 1.$

Illustrative Gaussian example.

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Deterministic transition

$$Y = \frac{X}{2}.$$

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Variance: $\frac{4}{16} + \frac{3}{4} = 1.$

Deterministic transition

$$Y = \frac{X}{2}.$$

Variance: $\frac{4}{4} = 1.$

Both give the same distribution of Y, but different joint distributions of (X, Y).

The deterministic transition gets all its randomness from X.

Simply deleting Z from the first rule gives $Y = X/4 \sim \mathcal{N}(0, 1/4)$: the wrong variance.

Illustrative Gaussian example.

DDIM: Deterministic Sampling

Sampling (DDIM)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$; choose a grid $T = t_K > \cdots > t_0 = 0$.
2. For $k = K, \ldots, 1$, set $t = t_k$, $s = t_{k-1}$ and compute:

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) \qquad \text{predict noise,}$$

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\, \hat{\boldsymbol{\epsilon}}}{\sqrt{\bar{\alpha}_t}} \qquad \text{estimate clean data,}$$

$$\mathbf{x}_s = \sqrt{\bar{\alpha}_s}\, \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_s}\, \hat{\boldsymbol{\epsilon}} \qquad \text{update state.}$$

3. Return $\mathbf{x}_0$.

DDIM: Deterministic Sampling

Sampling (DDIM)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$; choose a grid $T = t_K > \cdots > t_0 = 0$.
2. For $k = K, \ldots, 1$, set $t = t_k$, $s = t_{k-1}$ and compute:

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Fewer steps: choose $K \ll T$; K steps require K network evaluations and no retraining.

Outline

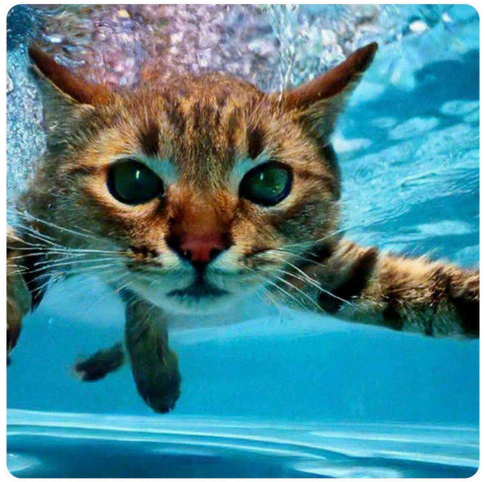
01 DDPM as a Score-Based Generative Model

02 **Model Guidance**

Classifier Guidance
Classifier-Free Guidance

Guidance

- Up to now, we have focused on **unconditional** generative models $p_{\theta}(\mathbf{x})$.
- In practice, most generative models are **conditional** (in the diffusion era it is called guided): $p_{\theta}(\mathbf{x}|\mathbf{y})$.
- Here, $\mathbf{y}$ might denote a class label or **text** (as in text-to-image tasks).



Кот ныряет в бассейн, как ребенок на обложке альбома Nevermind, реалистично



рука человека с пятью пальцами, ни четыремя, ни шестью, а с 5 (пять) пальцами

Conditional Models

In practice, we're typically interested in learning conditional models (sampling from conditional distribution $p_{\text{data}}(\mathbf{x}|\mathbf{y})$).

- $\mathbf{y} = \emptyset$, $\mathbf{x}$ = image $\Rightarrow$ unconditional image model
- $\mathbf{y}$ = class label, $\mathbf{x}$ = image $\Rightarrow$ class-conditional image model
- $\mathbf{y}$ = text prompt, $\mathbf{x}$ = image $\Rightarrow$ text-to-image model
- $\mathbf{y}$ = image, $\mathbf{x}$ = image $\Rightarrow$ image-to-image model
- $\mathbf{y}$ = image, $\mathbf{x}$ = text $\Rightarrow$ image-to-text (image captioning) model
- $\mathbf{y}$ = English text, $\mathbf{x}$ = Russian text $\Rightarrow$ sequence-to-sequence model (machine translation)
- $\mathbf{y}$ = sound, $\mathbf{x}$ = text $\Rightarrow$ speech-to-text (automatic speech recognition) model
- $\mathbf{y}$ = text, $\mathbf{x}$ = sound $\Rightarrow$ text-to-speech model

Label Guidance

Label: Ostrich (10th ImageNet class)



VQ-VAE (Proposed) BigGAN deep

Razavi A., Oord A., et al. Generating Diverse High-Fidelity Images with VQ-VAE-2, 2019

Text Guidance

Prompt: Anna
SDXL fine-tuned on Disney princesses. Same initial noise; only γ changes.

$\gamma = 3.5$ $\gamma = 4.5$ $\gamma = 5.5$ $\gamma = 6.5$



Kasymov A., et al. AutoLoRA: AutoGuidance Meets Low-Rank Adaptation for Diffusion Models, 2024. Fig. 3, top row (selected columns). CC BY 4.0.

Guidance in Generative Models

How to make a guided model?

Instead of sampling from $p_{\boldsymbol{\theta}}(\mathbf{x})$, we sample from $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{y})$.

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- Encoder $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}, \mathbf{y})$ and decoder $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}, \mathbf{y})$ for VAEs;

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- $G_{\boldsymbol{\theta}}(\mathbf{z}, \mathbf{y})$ for NFs and GANs;
- $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$ for DDPMs.

Guidance in Generative Models

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- $G_{\boldsymbol{\theta}}(\mathbf{z}, \mathbf{y})$ for NFs and GANs;
- $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$ for DDPMs.

Challenge

- Empirically, images sampled with this procedure do not fit well enough to the desired label $\mathbf{y}$.
- Being able to control the strength of guidance is especially valuable.

Outline

01 DDPM as a Score-Based Generative Model

02 **Model Guidance**

Classifier Guidance

Classifier-Free Guidance

Classifier Guidance

DDPM Sampling

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise the image (unconditional generation):

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}$$

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Classifier Guidance

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Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) + \sigma_t \cdot \boldsymbol{\epsilon}$$

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What is the link between $\nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t)$ and $\nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y})$?

Classifier Guidance: Guided Score Function

Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) + \sigma_t \cdot \boldsymbol{\epsilon}$$

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$$\nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log \left(\frac{p_{\boldsymbol{\theta}}(\mathbf{x}_t) p(\mathbf{y}|\mathbf{x}_t)}{p(\mathbf{y})} \right)$$

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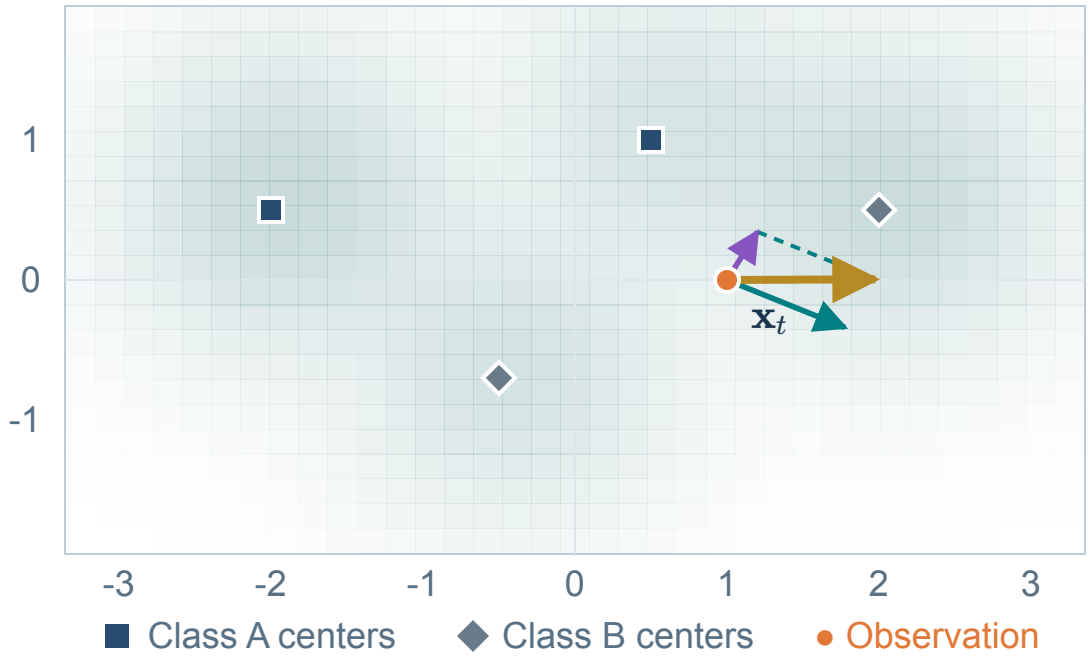
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Classifier Guidance: Which Way Does It Push?

Four Gaussian components, two classes, one fixed noise level.

Fixed observation · target class B · γ = 3.



Gaussian component centers and the marginal density.

Unconditional score

$\mathbf{s}_t(\mathbf{x}_t)$ (1.08, 1.85)

Classifier contribution

$\gamma \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$ (4.18, -1.83)

Guided score

$\mathbf{s}_t^{\gamma}(\mathbf{x}_t, \mathbf{y})$ (5.25, 0.03)

$p(\mathbf{y}|\mathbf{x}_t)$ = 0.648

Exact toy scores at a fixed noise level. At this point, the common scale is fixed across γ and classes.

Guidance adds a class-directed correction to the unconditional score.

Classifier Guidance: Guidance Scale

Guided Score Function

$$\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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- Let us assume $\mathbf{y}$ is a class label.
- $p(\mathbf{y}|\mathbf{x}_t)$ is a classifier for noisy inputs.
- $p(\mathbf{y}|\mathbf{x}_t)$ is responsible for model guidance.

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Guidance Scale

It is a natural idea to scale up the contribution of the guidance

$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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- The guidance scale γ adjusts the strength of classifier guidance.
- $\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$ is not the true guided score function $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$.

Classifier Guidance: Distribution Sharpening

Scaled Guided Score Function

$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

Dhariwal P., Nichol A. Diffusion Models Beat GANs on Image Synthesis, 2021
Sadat S. et al. Eliminating Oversaturation and Artifacts of High Guidance Scales in Diffusion Models, 2025

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Scaled Conditional Distribution at a Fixed Noise Level

$$\nabla^{\gamma}_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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&= \nabla_{\mathbf{x}_t} \log \left(\frac{p_{\boldsymbol{\theta}}(\mathbf{x}_t) p(\mathbf{y}|\mathbf{x}_t)^{\gamma}}{Z} \right)
\end{aligned}$$

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\end{aligned}$$

Note: Increasing γ sharpens $p(\mathbf{y}|\mathbf{x}_t)$, increasing the contrast

$$\hat{p}(\mathbf{y}|\mathbf{x}_t) \propto p(\mathbf{y}|\mathbf{x}_t)^{\gamma}.$$

Very large γ can reduce diversity and introduce artifacts.

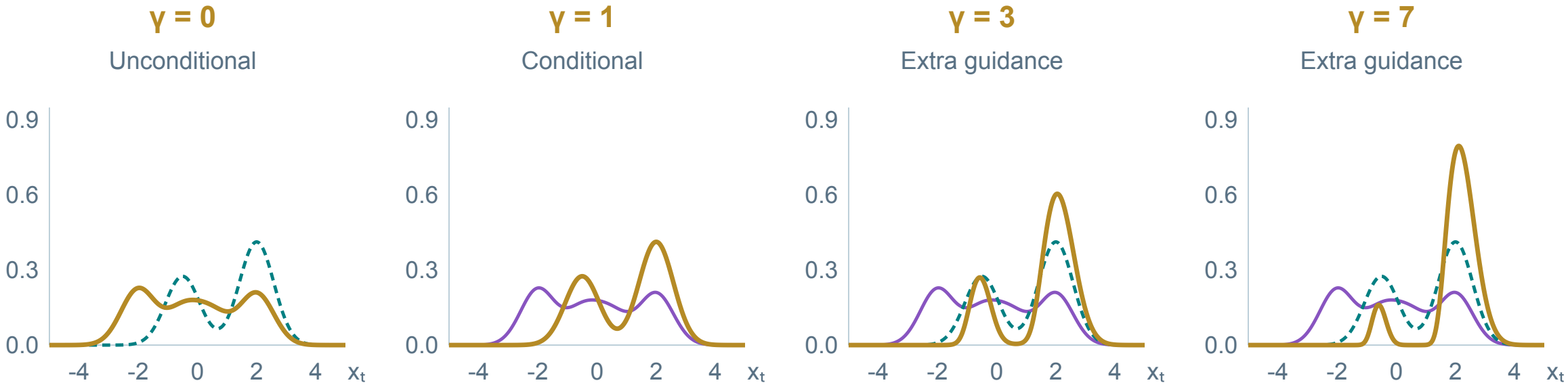
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Guidance Scale Changes the Density

$$q_\gamma(x_t|y) \propto q(x_t)\, p(y|x_t)^\gamma \qquad \text{at a fixed noise level.}$$

Same one-dimensional toy and target class B; only γ changes.

$q(x_t)$ $q(x_t|y)$ · dashed $q_\gamma(x_t|y)$



Increasing γ shifts mass toward regions with higher classifier confidence.

A 1D version of the Gaussian-mixture toy, at a fixed noise level. This plot describes a density at that level.

Classifier Guidance: Overview

Training

1. Train the DDPM as before.
2. Train an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ on noisy data (time-dependent).

Classifier Guidance: Overview

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Sampling (Guided)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise with the scaled guided score function:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \boldsymbol{\epsilon}.$$

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Classifier-Free Guidance

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- The previous approach relies on training an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ for noisy images.
- We now introduce a method to sidestep this requirement.

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$$\nabla^{\gamma}_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

Bayes theorem

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log \left(\frac{p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) p(\mathbf{y})}{p_{\boldsymbol{\theta}}(\mathbf{x}_t)} \right)$$

Classifier-Free Guidance

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Classifier-Free Guidance

Scaled Guided Score Function

$$\nabla^{\gamma}_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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\end{aligned}
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Classifier-Free Guidance: Formulation

Scaled Guided Score Function

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Naive training approach

- Train an unguided score function model $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t)$.
- Train a guided score function model $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$.

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- Use their affine combination at inference.

Sampling (Guided)

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \boldsymbol{\epsilon}$$

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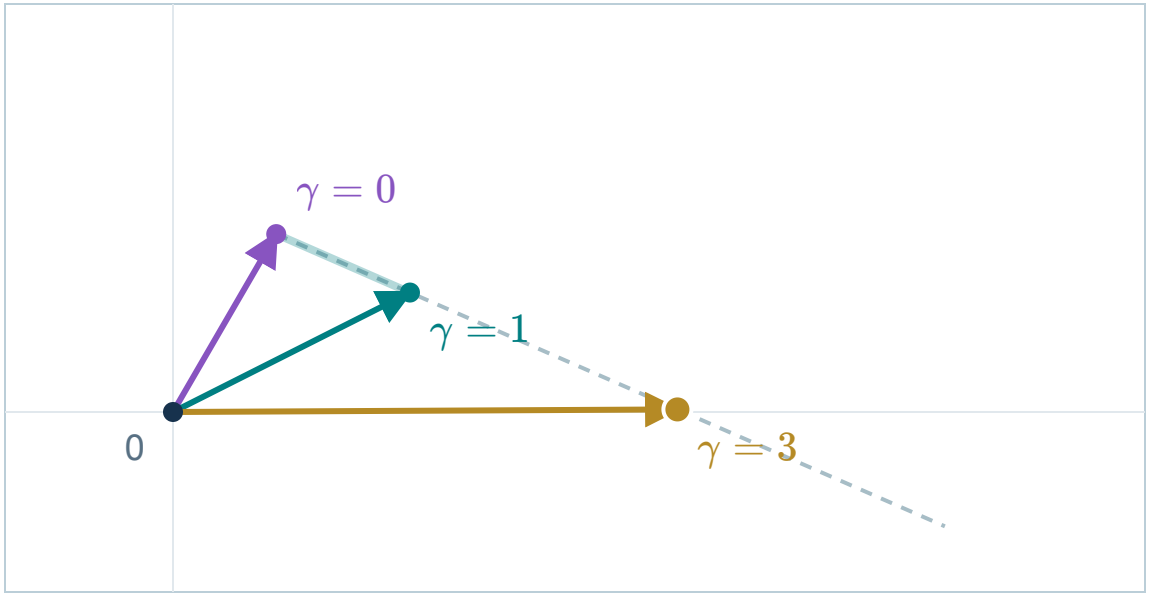
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How to avoid training two separate score function models?

CFG: Interpolation and Extrapolation

$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t} = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing) + \gamma\big(\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) - \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing)\big)$$

Same two scores: $\gamma = 0$, $\gamma = 1$ and $\gamma = 3$.



Score space · endpoints of vectors

The two base scores are fixed; γ moves the guided endpoint along their line.

$\gamma = 3.0$ · Extrapolation

$\gamma = 0$

$\mathbf{s}^0_t = \mathbf{s}_t(\mathbf{x}_t, \varnothing)$

$\gamma = 1$

$\mathbf{s}^1_t = \mathbf{s}_t(\mathbf{x}_t, \mathbf{y})$

$\gamma > 1$

Continue beyond the conditional endpoint.

$\varnothing$ denotes no condition. Analytic toy scores at one fixed observation and noise level.

For $\gamma > 1$, CFG extrapolates beyond the conditional score.

Classifier-Free Guidance

$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = (1-\gamma) \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$$

CFG

1. Introduce the "absence of conditioning" label $\mathbf{y} = \varnothing$.
2. Identify the unguided score function $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing)$.
3. Train a single model $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$ on **supervised** data, dropping the label $\mathbf{y}$ with some fixed probability (simulating $\mathbf{y} = \varnothing$).
4. At inference, evaluate the model twice to obtain $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing)$ and $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$.

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$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = (1-\gamma) \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$$

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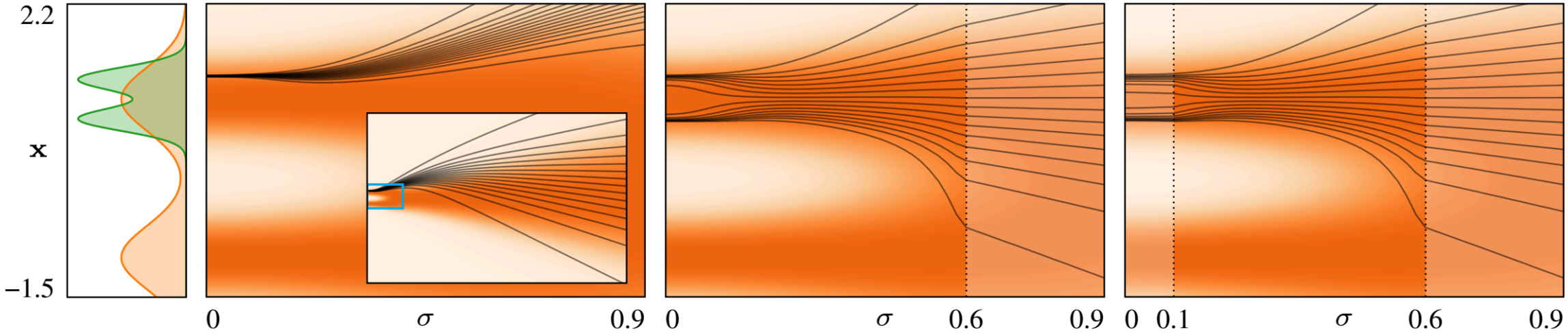
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Classifier-Free Guidance: Guidance Interval

Apply extra guidance only at intermediate noise levels.

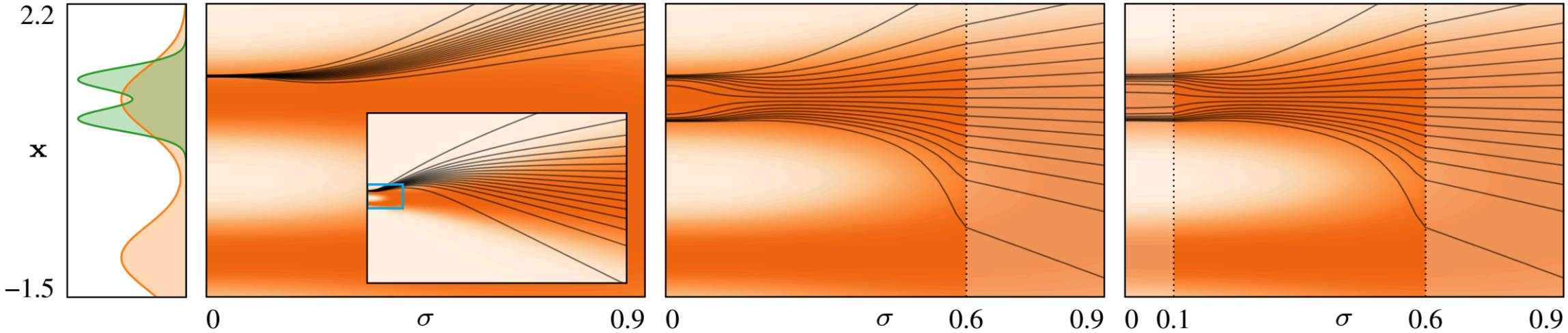


(a) PDFs (b) Guidance everywhere (c) Guidance at $\sigma \in (0, 0.6]$ (d) Guidance at $\sigma \in (0.1, 0.6]$

Kynkäänniemi T. et al. Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models, 2024. Fig. 2.

Classifier-Free Guidance: Guidance Interval

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(a) PDFs (b) Guidance everywhere (c) Guidance at $\sigma \in (0, 0.6]$ (d) Guidance at $\sigma \in (0.1, 0.6]$

Toy example ($\gamma = 6$): guidance everywhere drops a mode; limiting the interval restores both.

Outside the interval, $\gamma_t = 1$: **conditioning remains**. Sampling runs **right to left** ($\sigma \downarrow$).

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Classifier-Free Guidance: Guidance Distillation

Learn the guided prediction in a single model evaluation.

Teacher: two evaluations

$$\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing), \quad \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$$

Combine predictions using γ.

Meng C. et al. On Distillation of Guided Diffusion Models, 2023.
Black Forest Labs. FLUX.2 [dev] model card: guidance distillation.

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distill $\longrightarrow$

Student: one evaluation

$$\mathbf{s}_{\boldsymbol{\phi},t}(\mathbf{x}_t, \mathbf{y}, \gamma)$$

Pass γ as an extra input.

$$\mathbf{s}_{\boldsymbol{\phi},t}(\mathbf{x}_t, \mathbf{y}, \gamma) \approx (1-\gamma)\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing) + \gamma \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$$

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Training

1. Sample $(\mathbf{x}_0, \mathbf{y}) \sim p_{\text{data}}$, $t \sim \text{Uniform}\{1, \dots, T\}$, $\gamma \sim \text{Uniform}[\gamma_{\min}, \gamma_{\max}]$.
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Sampling

1. Evaluate the student once per step; keep the sampler and step count. Used in **FLUX.2 [dev]**.

Meng C. et al. On Distillation of Guided Diffusion Models, 2023.
Black Forest Labs. FLUX.2 [dev] model card: guidance distillation.

Summary

- DDPM and NCSN are connected through score matching and a rescaling of their Gaussian perturbations.
- DDPM and NCSN use different samplers; DDIM reuses the DDPM network on a shorter sampling grid.
- Guidance makes generation controllable through labels or text prompts.
- Classifier guidance turns an unconditional model into a conditional one by training an auxiliary classifier on noisy data.
- Classifier-free guidance needs no auxiliary classifier; guidance intervals can improve its quality-diversity trade-off, and distillation reduces evaluations per step.